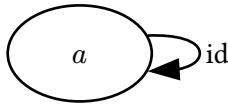


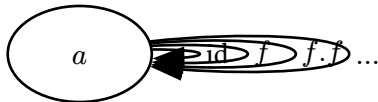
Chapter 3

1. Free construction:

A graph with one node and no edges

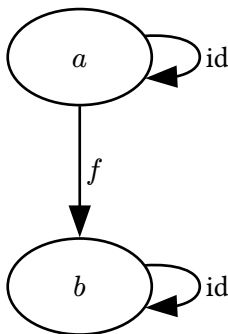


A graph with one node and one (directed) edge (hint: this edge can be composed with itself)



$\text{hom-set}(a, a) = \{\text{id}, f, f \cdot f, f \cdot f \cdot f, \dots\}$ (monoid where f is the generator)

A graph with two nodes and a single arrow between them



A graph with a single node and 26 arrows marked with the letters of the alphabet: a, b, c ... z.

Similar to the previous monoid but now with more possible combinations of arrows.

$\text{hom-set} = \{a..z\}^*$ where $ab = \text{"a after b"}$. Isomorphic to all possible strings. I.e. composition = string concatenation.

2. Orders

A set of sets with the inclusion relation: A is included in B if every element of A is also an element of B.

$$A \in B \wedge B \in A \Rightarrow A \subseteq B \wedge B \subseteq A \Rightarrow A = B$$

So when both are contained in one another, we get equality. If exactly one is contained in the other, we have a ordering between them. But if we have two sets, e.g. $\{0\}$ and $\{1\}$, that don't contain each other, we cannot compare them! So this is a *partial* order.

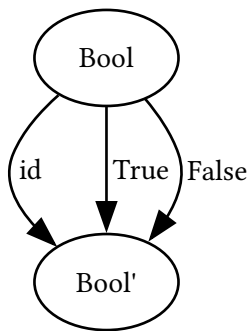
C++ types with the following subtyping relation: T1 is a subtype of T2 if a pointer to T1 can be passed to a function that expects a pointer to T2 without triggering a compilation error.

partial order. Trivially reflexive. Equivalent to extending a class. So antisymmetric also trivial (A extends B and B extends $A \Rightarrow A = B$). Transitive also true probably.

3. Considering that Bool is a set of two values True and False, show that it forms two (set-theoretical) monoids with respect to, respectively, operator && (AND) and || (OR).

$(a \ \&\& \ b) \ \&\& \ c = a \ \&\& \ (b \ \&\& \ c)$ (can be shown with truth table). Similar for $\parallel$. Identity for $\&\&$ is true: $a \ \&\& \ \text{true} = a = \text{true} \ \&\& \ a$. Identity for $\parallel$ is false.

4. Represent the Bool monoid with the AND operator as a category: List the morphisms and their rules of composition.



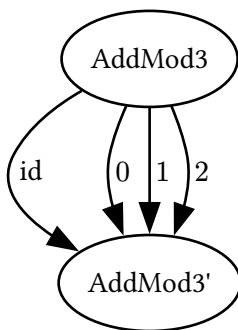
(Bool = Bool', they are drawn as different nodes for readability but represent the same object).

- $\text{True} \ \&\& \ \text{True} = \text{True}$
- $\text{True} \ \&\& \ \text{False} = \text{False} \ \&\& \ \text{True} = \text{False} \ \&\& \ \text{False} = \text{False}$.
- $\text{True} \ \&\& \ \text{id} = \text{True}$
- $\text{False} \ \&\& \ \text{id} = \text{False}$

Note: $\text{id} = \text{True}$ since:

- $\text{True} \ \&\& \ \text{id} = \text{True} \ \&\& \ \text{True} == \text{True}$
- $\text{False} \ \&\& \ \text{id} = \text{False} \ \&\& \ \text{True} == \text{False}$

5. Represent addition modulo 3 as a monoid category.



- $n + m = n + m \% 3$.
- E.g.
 - $0 + 1 = 1$
 - $2 + 1 = 0$.

$\text{id} = 0$ inverse of $1 = 2$ So this is an isomorphism??? Yes! Isomorphism mapping:

- $0 \rightarrow 0$
- $1 \rightarrow 2$
- $2 \rightarrow 1$